

Literature Status for a Schlumprecht Ergodicity Question in arXiv:math/0511170

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Status

This note records a literature-implied answer to a concrete question in Ferenczi, arXiv:math/0511170, and a partial later-literature answer to the broader ergodicity conjecture stated there.

Source Signals

In the introduction, the source paper states the conjecture that every separable Banach space is either isomorphic to ℓ_2 or ergodic.

Later, after proving that E_0 Borel reduces to permutative equivalence on block bases of Schlumprecht's space S , the source says:

It is however still unknown if S is ergodic.

Supporting Result

The later paper Cuellar Carrera, arXiv:1611.05500, proves that every separable non-near-Hilbert Banach space is ergodic. Equivalently, a non-ergodic separable Banach space must be near Hilbert. This narrows the broad conjecture from arXiv:math/0511170 to the near-Hilbert case.

The same later paper proves that the Schlumprecht-type spaces $S_{p,r}$ and their duals are not near Hilbert, and explicitly concludes that they are ergodic. Since Schlumprecht's original space is $S_{1,\infty}$, this gives a positive answer to the source paper's concrete question: Schlumprecht's space S is ergodic.

Scope

This packet is not a new theorem. It records an already-known later implication. The broad conjecture that ℓ_2 is the only non-ergodic separable Banach space remains open in the near-Hilbert range. The Schlumprecht-space question from the source paper is settled positively by the later theorem.

References

1. V. Ferenczi, *On the number of non permutatively equivalent sequences in a Banach space*, arXiv:math/0511170.

2. W. Cuellar Carrera, *Non-ergodic Banach spaces are near Hilbert*, arXiv:1611.05500.